

B.E. I YEAR EXAMINATION, JAN 2024
(COMPUTER ENGINEERING – A & B)
1MERC3--ELEMENTS OF MECHANICAL ENGINEERING

2024250

Duration: 3 Hrs.

Maximum Marks: 60

Note: All Questions are compulsory to attempt. Attempt any two parts within each Question.
Use of Steam tables are permitted in the examination.

Q1 (a) A partition wall divides a rigid vessel containing air into compartments, the volume of each being 0.2 m^3 . The pressure of air in one of the compartment is maintained at 2 bar and in other compartment is 4 bar. Initial temperature of air of both the compartment is the same. The vessel is heated and 125 kJ of heat is supplied resulting puncture of the partition wall. Calculate the final pressure of air when the equilibrium is attained. 06

(b) Explain throttling process and prove enthalpy before throttling is equal to enthalpy after throttling. 06

(c) In a vessel, 10 kg of O_2 is heated in a reversible, non-flow, constant volume process, so that the pressure of O_2 is increased two times that of the initial value. The initial temperature is 20°C . Calculate (a) The final temperature (b) The change in internal energy (c) The change in enthalpy (d) The heat transfer. Take $R=0.259 \text{ KJ/Kg K}$ and $C_v=0.652 \text{ KJ/Kg K}$ for oxygen. 06

Q2 (a) A spherical shell of 30 cm in radius contains saturated steam and water at 300°C . Calculate the mass of each if their volumes are equal. 06

(b) Steam at 10 bar and 200°C is cooled till it becomes dry saturated and is then throttled to 1 bar pressure. Determine the change in enthalpy and heat transferred during each process. Also calculate the quality of steam at the end of throttling process 06

(c) What is meant by quality of steam? Describe with a neat sketch the working of a combined separating and throttling calorimeter. And derive the quality of steam measured by this calorimeter. 06

Q3 (a) Show that except Monoatomic gases all other gases has an increase in specific heats at high temperatures and Derive the variation in air standard efficiency of diesel cycle with variable specific heats. 06

(b) Explain and describe the construction and working of a carburetted single cylinder 4 stroke petrol engine. 06

(c) A diesel engine operating on air standard diesel cycle has 20 cm bore and 30 cm stroke. The clearance volume is 420 cm^3 . The fuel is injected at constant pressure for 5% of the stroke. Calculate the air standard efficiency. If the cut off is delayed from 5 to 8%. What will be the percentage loss in efficiency? In both cases, the compression ratio is same. 06

- Q4 (a) Explain in detail low pressure gas welding plant. And explain types of gas flame. 06
- (b) Describe TIG and MIG welding processes. 06
- (c) Describe the Submerged Arc Welding Process with neat sketch. 06
- Q5 (a) what are principle ingredients of moulding sand. Explain properties of moulding sand. 06
- (b) Define pattern allowance. List pattern allowances. Explain shrinkage and distortion allowance. 06
- (c) Explain briefly the shell Mould casting process. Write its process characteristics and applications 06